AI.102

Robotics Fundamentals

COURSE OVERVIEW

Core robotics concepts: robot anatomy, kinematics basics, sensors, actuators, and the integrate sense-plan-act cycle.

1

SESSION

60 MINUTES

ORIENTATION: THE ROBOT AS A SYSTEM

KEY TOPICS

- What defines a robot
- Sense–plan–act cycle
- Robot vs. automaton
- Applications across sectors

TEACHING FOCUS

Establish a unified view of a robot as an integrated system.

KEY TAKEAWAY A robot senses, plans, and acts in the physical world.

2

SESSION

60 MINUTES

ROBOT ANATOMY & CLASSIFICATION

KEY TOPICS

- Links, joints, and degrees of freedom
- Serial vs. parallel manipulators
- Mobile, humanoid, and industrial robots
- Specification and workspaces

TEACHING FOCUS

Classify robots by structure and intended workspace.

KEY TAKEAWAY Structure determines capability.

3

SESSION

60 MINUTES

ACTUATORS & DRIVE SYSTEMS

KEY TOPICS

- Electric motors: DC, stepper, servo
- Hydraulic and pneumatic actuators
- Transmission: gears, belts, harmonic drives
- Actuator selection criteria

TEACHING FOCUS

Match actuators to motion and force requirements.

KEY TAKEAWAY Actuators are the muscles of a robot.

4

SESSION

60 MINUTES

SENSORS & PERCEPTION BASICS

KEY TOPICS

- Proprioceptive vs. exteroceptive sensors
- Encoders, IMUs, LiDAR, cameras
- Sensor characteristics: noise, range, resolution
- Sensor fusion introduction

TEACHING FOCUS

Choose and interpret the robot's senses.

KEY TAKEAWAY Sensors convert the physical world into data.

5

SESSION

60 MINUTES

CONTROL BASICS FOR ROBOTICS

KEY TOPICS

- Open-loop vs. closed-loop control
- Feedback and error signals
- P, PI, PID concepts
- Set-point tracking in motion

TEACHING FOCUS

Regulate robot motion with feedback control.

KEY TAKEAWAY Feedback closes the sense–plan–act loop.

6

SESSION

60 MINUTES

INTEGRATION, SAFETY & REVIEW

KEY TOPICS

- Hardware–software integration
- Robot safety standards and E-stops
- Course recap and assessment
- Path toward robotics design

TEACHING FOCUS

Integrate knowledge with safety-first engineering practice.

KEY TAKEAWAY Safe integration is the mark of a professional.